Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [18SE5DE12]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

SOFTWARE ARCHITECTURE AND DESIGN

Time: 3 Hours **Maximum Marks: 70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.

1.	**a.**	Find a commonly accepted definition of ISO/IEC/IEEE 42010 system architecture and discuss what it has in common with software architecture.	**07**
	b.	Software architecture brings a number of tangible benefits. Discuss	**07**
2.	**a.**	Compare and Contrast the various types of architects based on their distinguishing factors, strength and weakness.	**07**
	b.	Draw and explain the difference between agile and waterfall model while implementing in small projects.	**07**
3.	**a.**	Summarize the foundation of understanding a domain that includes general business knowledge and understanding of your organization's business.	**07**
	b.	Analyze on the concept of Domain-Driven Design.	**07**
		OR	
4.	**a.**	Discuss how the concept of a ubiquitous language is used to mitigate the Risks and how it can be used in Modeling?	**14**
5.	a.	Define Quality attributes. Determine and prepare the test cases for Usability Testing.	**07**
	b.	Compare and contrast terms errors, faults and failures using suitable examples.	**07**
		OR	
6.	**a.**	Demonstrate the availability based on time and request success rate with example.	**07**
	b.	Suppose you are testing defect coin problem artifacts, Identify the various fault recovery strategies. What steps could have been taken to prevent the various classes of faults?	**07**
7.	**a.**	Elucidate various modularity and control system commonly used on any organizational module.	**07**
	b.	Define Modularity. Describe the various coupling methods used in software design.	**07**

OR

8.	**a.**	Software architecture is often compared to the architecture of buildings as a conceptual analogy. What are the strong points of that analogy? What is the correspondence in buildings to software architecture structures and views and patterns? What makes a good architecture?	**07**
	b.	'A system must be loosely coupled and highly cohesive'. Justify.	**07**